Andrea Valla

Software Architect & Principal-Level Full-Stack Engineer | Industrial Systems, Distributed Software & AI Agents

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PROFESSIONAL SUMMARY

Software architect and hands-on full-stack engineer with 20+ years of experience designing, building, and operating production systems. Independently architected the foundation of Agile Factory, a modular industrial MES built for customer-specific deployments, real-time shop-floor workflows, and equipment/ERP integrations. More recently, engineered PostgreSQL-backed ecommerce applications and AI agent orchestration infrastructure. Focused on pragmatic architecture, reliability, extensibility, and long-term maintainability.

PROFESSIONAL EXPERIENCE

Solution Architect, Full-Stack Developer & System Administrator | BrandsDistribution.com

Feb 2022 - Sep 2026 | Turin, Italy

- Built queue-backed BDroppy integrations with Shopify, Squarespace, Wix, and EKM supporting 50,000+ connected stores; shipped apps through the Shopify and Wix app marketplaces.
- Designed and built AutoEpoque, a classic-car marketplace for classifieds and auctions using React, TypeScript, Supabase/PostgreSQL, Row-Level Security (RLS), BullMQ/Redis background jobs, and pgTAP database tests.
- Architected and built Aidify, an AI-powered chatbot platform integrated with ecommerce channels.
- Built a BI and analytics platform using BigQuery, Airbyte, Cube.js, and Looker Studio to support cost and ROI analysis.
- Owned software architecture, backend, frontend, integrations, data, and production operations across multiple products.

Technologies: TypeScript, React, Node.js, Bun, PostgreSQL, Supabase, Redis, BigQuery, GraphQL

CTO & Co-founder | Agile Factory

Mar 2018 - Jan 2022 | Turin, Italy

- Co-founded Agile Factory and independently designed and implemented its foundational MES architecture, from domain models and production workflows to customer deployments and operations.
- Created a modular Node.js monorepo, separating domain models, business services and GraphQL APIs, with a lightweight plugin/hook system to extend customer-specific behavior without forking the core.
- Built the React Manager and shop-floor HMI foundations, GraphQL subscriptions and schema-declared API permissions; developed integration paths for industrial equipment and ERP/ETL systems using OPC UA and MQTT.
- Used Cypress end-to-end workflows to verify application behavior and generate screenshots for user documentation in the same process.
- Designed the original platform when it served approximately 5 customers; it subsequently grew to approximately 50 customers by 2026 under a different development team.

Technologies: Node.js, React, GraphQL, Apollo, MongoDB, Redis, Moleculer, Docker, Cypress, OPC UA, MQTT

Full-Stack Developer | Industrial Cloud

Jun 2015 - Mar 2018 | Turin, Italy

- Delivered manufacturing web applications from analysis and product requirements through implementation, customer deployment, and ongoing updates.

Technologies: Node.js, React, Elasticsearch, MySQL, PHP

Full-Stack Developer | Intesa Sanpaolo

Sep 2010 - Jun 2015 | Moncalieri, Italy

- Built business applications, ETL/data workflows, and reporting systems using SQL Server, SSIS/SSRS, .NET, and JavaScript.

Developer & System Administrator | Reply S.p.A.

Sep 2003 - Sep 2010 | Turin, Italy

- Developed software and administered Linux, Unix, and Microsoft systems, including constrained enterprise infrastructure and production-risk containment.

Unix System Administrator | Global Value Services

Jul 2002 - Jul 2003 | Turin, Italy

- Managed systems and applications supporting FCA web properties in HP-UX environments.

SELECTED PROJECTS

Agile Factory | Co-founder, CTO & original software architect

2018 - 2022 | https://www.agilefactory.it

- Independently designed the foundations of a modular Manufacturing Execution System, including production orders, operations, operator-facing HMI workflows, and customer-specific deployment patterns.
- Built reusable Node.js business modules, GraphQL APIs with declarative authorization and real-time subscriptions, and a lightweight plugin system for industrial integrations and extensions.
- Combined Cypress end-to-end validation with screenshot generation for product manuals, reducing duplicated testing and documentation work.
- The platform grew from approximately 5 customers at my departure in 2022 to approximately 50 by 2026, continuing to evolve under a different development team. Customer counts are approximate.

Technologies: Node.js, React, GraphQL, MongoDB, Redis, Cypress, OPC UA, MQTT

AI Office | Creator

2026 - Present | https://github.com/avalla/ai-office

- Built an open-source multi-agent orchestration platform for AI agents, workflows, tasks, pipelines, and durable executions.
- Implemented concurrency control, task locking, LLM/model routing, audit logs, tool permissions, and governance.
- Integrated OpenAI, Codex, and Claude Code behind provider-neutral interfaces.
- Designed pluggable persistence and knowledge layers using PostgreSQL, SQLite, and SurrealDB, with extensible agents and domain behavior.

Technologies: TypeScript, Bun, PostgreSQL, SQLite, SurrealDB, OpenAI, Codex, Claude Code

AutoEpoque | Product architecture & engineering at BrandsDistribution

2025 - Present | https://autoepoque.com

- Own end-to-end architecture and implementation for a classic-car marketplace supporting classifieds and auctions.
- Built a React/TypeScript monorepo on Supabase/PostgreSQL with Row-Level Security (RLS), BullMQ/Redis jobs, media workflows, internationalization, and pgTAP database and security tests.

Technologies: TypeScript, React, PostgreSQL, Supabase, Redis, BullMQ, pgTAP

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript, C#, PHP, SQL
Frontend: React, React Native, Apollo, URQL, Redux, Vite, Tailwind CSS
Backend & APIs: Node.js, Bun, Deno, Express, .NET, GraphQL, REST APIs
Databases & Data: PostgreSQL, Supabase, Redis, MongoDB, SQL Server, MySQL, BigQuery, Airbyte, Cube.js
Architecture: Software Architecture, System Design, Distributed Systems, API Integration, Third-Party Integrations, Background Jobs, Event-Driven Systems
DevOps & Cloud: Linux/Unix, Docker, GitHub Actions, AWS, GCP, Azure, Cloudflare, CI/CD
Testing: Automated Testing, Vitest, pgTAP, Database & RLS Testing
AI / LLM: LLMs, AI Agents, Agent Orchestration, OpenAI API, LangChain, Model Routing, LLM Gateways, Usage/Cost Metering, Tool Permissions
Industrial: Manufacturing Execution Systems (MES), IIoT, OPC UA, MQTT, ERP Integration, ETL

LANGUAGES

Italian (Native) | English (Fluent) | German (Basic)